MSC THESIS DEFENSE

# Data-Driven Cognitive Phenotyping in Acquired Brain Injury

*Do routine neuropsychological data reveal cognitive subtypes?*

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WHY THIS MATTERS

# The clinical problem: heterogeneity

- ABI can disrupt attention, memory, language, executive function, orientation and visuospatial cognition.
- Impairment does not follow the same pattern in every patient.
- Diagnosis and injury severity are useful categories — but they do not capture cognitive heterogeneity.
- Incomplete assessments are common, especially among more impaired patients.

**Research question**

*Can data-driven clustering recover clinically meaningful cognitive organisation when missing data are handled explicitly — rather than ignored?*

POSITIONING

# Research gap and contribution

| Common limitation in prior work | This dissertation |
| --- | --- |
| Samples often below 200 participants | 17,406 assessments · 7,285 patients |
| Complete-case analysis | Ten imputation strategies compared |
| Limited algorithmic sensitivity testing | Hyperparameter, seed and missingness checks |
| Predetermined number of clusters | Density-based clustering (HDBSCAN) |
| Limited clinical validation | Association with clinical-unit assignment |

HYPOTHESES

# Four hypotheses, objective criteria

**H1** — **Are stable clusters recoverable?**

At least eight methods meet selected embedding thresholds; discreteness requires separate evidence

`Silhouette ≥ 0.40 · noise ≤ 30% · ≥ 2 clusters`

**H2** — **Robust to imputation choice?**

Assessments retain similar assignments across imputations under a fixed reference rule

`Median pairwise ARI > 0.50 (fixed reference)`

**H3** — **Are tiers associated with clinical units?**

Severity tier relates to clinical unit, even after accounting for diagnosis

`χ² test · Cramér's V > 0.10 · CMH adjust for diagnosis`

**H4** — **Does domain aggregation help?**

Domain aggregation improves conditioning while preserving similar separation

`Higher silhouette · lower VIF & condition number`

THE DATA

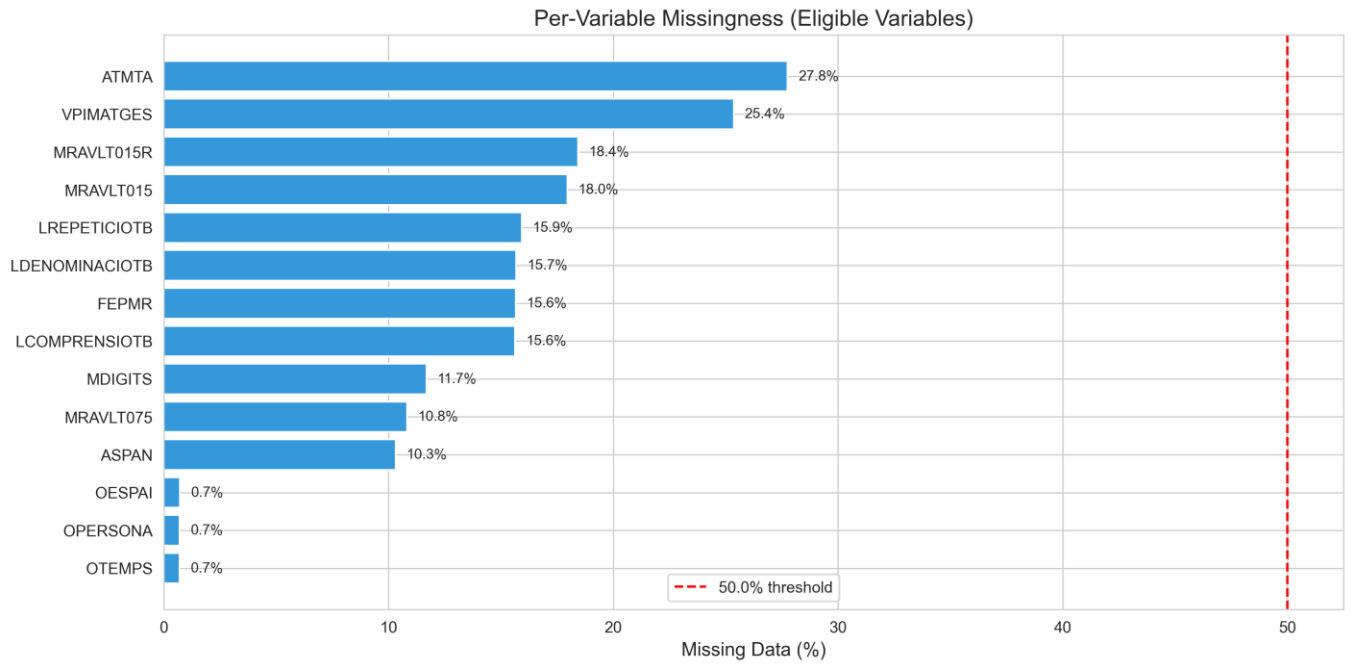
# A large but incomplete cohort

**17,406**
assessments

**7,285**
patients

**14**
variables

**6**
cognitive domains



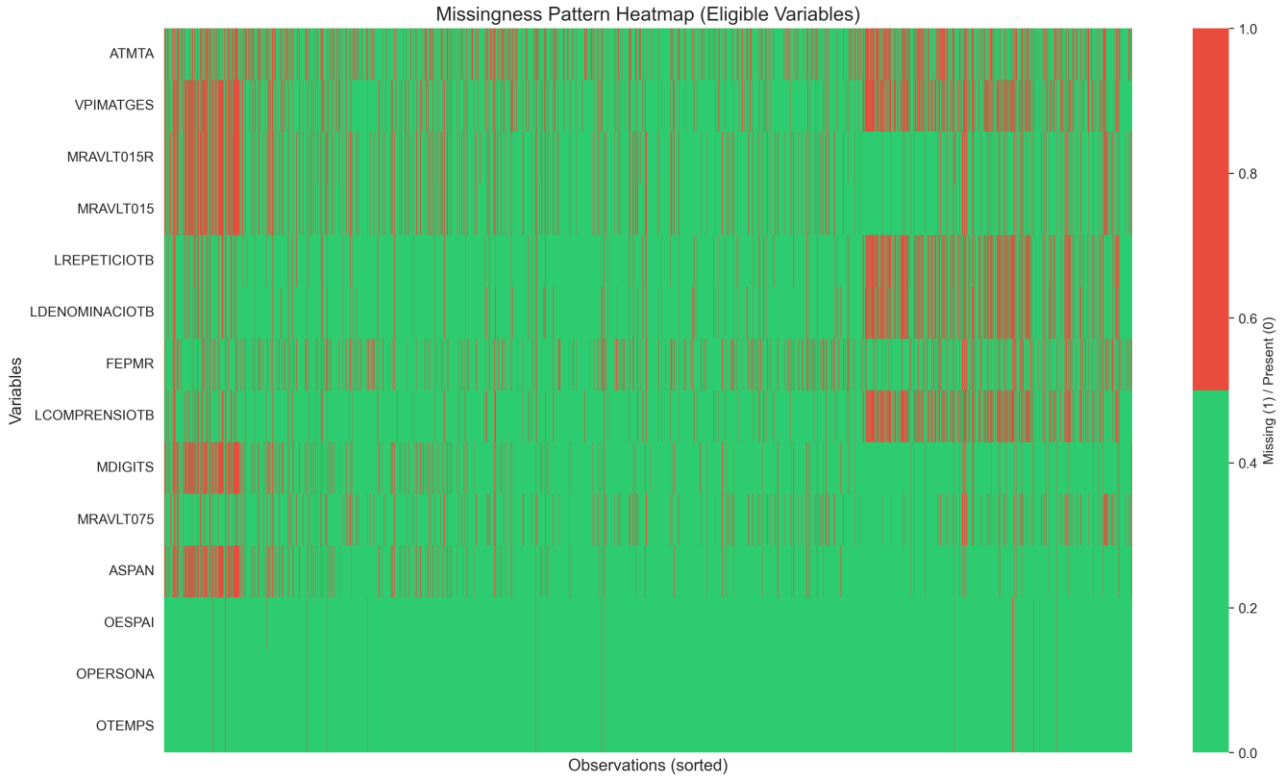

**53.1%**

of rows are complete across all 14 variables

- Complete-case analysis discards 46.9% of assessments.
- Missingness follows structured patterns; its mechanism remains uncertain.
- Imputation retains partially observed assessments.

METHOD RATIONALE

# Why imputation is necessary




## Missingness reflects the clinical testing process

- Fatigue or reduced motivation
- Cognitive overload or task aversion
- Disorientation or motor limitations
- Limited tolerance for long batteries
- Clinician selects a more suitable test

*Complete-case selection changes the cohort. Whether excluded assessments are more impaired cannot be established from missing scores alone.*

METHODS

# Analysis pipeline

**Impute**
10 strategies
simple → deep learning

→

**Scale**
orient scores —
higher = better

→

**Aggregate**
14 variables →
6 domains

→

**UMAP**
manifold
embedding

→

**HDBSCAN**
density clustering
+ sensitivity

## Fixed-reference design (H2)

A scaler, UMAP embedding and three-group k-means rule are fitted once on MICE domain scores. Other imputations use those same transformations and assignment rule.

✓ **Tests assignment agreement across imputations while the reference geometry is fixed.**

✗ **Not tested** that re-fitting the whole pipeline per dataset would rediscover the same cluster hierarchy.

# UMAP

*Uniform Manifold Approximation & Projection · McInnes, Healy & Melville, 2018*

- Builds a fuzzy k-NN graph of neighbourhood structure.
- Balances local and global geometry.
- Reduces dimensionality before density clustering.

*Caveat: separation in UMAP space can be optimistic — it must be backed by sensitivity analyses.*

## FORMAL DEFINITION

$$p_{j|i} = \exp\left( - \frac{d(x_i, x_j) - \rho_i}{\sigma_i} \right), \qquad \sum_j p_{j|i} = \log_2 k$$

$$p_{ij} = p_{j|i} + p_{i|j} - p_{j|i}\, p_{i|j}$$

$$q_{ij} = \left(1 + a \left\| y_i - y_j \right\|_2^{2b}\right)^{-1}$$

$$\mathcal{C} = \sum_{i \neq j} p_{ij} \log \frac{p_{ij}}{q_{ij}} + (1 - p_{ij}) \log \frac{1 - p_{ij}}{1 - q_{ij}}$$

METHODS · DENSITY CLUSTERING

# HDBSCAN

*Hierarchical Density-Based Spatial Clustering of Applications with Noise · Campello, Moulavi & Sander, 2013*

- Needs no predefined number of clusters.
- Finds regions dense across density levels.
- Labels uncertain points as noise, not forced.

*Minimum cluster size, density thresholds and selection settings affect the recovered partition.*

FORMAL DEFINITION

$$\mathrm{core}_k(x) = d\big(x,\, N_k(x)\big)$$

$$d_{\mathrm{mreach}}(a,b) = \max\{\mathrm{core}_k(a),\, \mathrm{core}_k(b),\, d(a,b)\}$$

$$S(C) = \sum_{x \in C} \big(\lambda_{\max}(x) - \lambda_{\mathrm{birth}}(C)\big), \qquad \lambda = \frac{1}{\mathrm{dist}}$$

$$\widehat{\mathcal{P}} = \arg\max_{\mathcal{P}} \ \textstyle\sum_{C \in \mathcal{P}} S(C) \quad \text{(non-nested)}$$

# Ten imputation strategies

**Simple → model-based**

1. Mean imputation
2. Expectation–maximization
3. MICE (reference)
4. K-nearest neighbours
5. MissForest

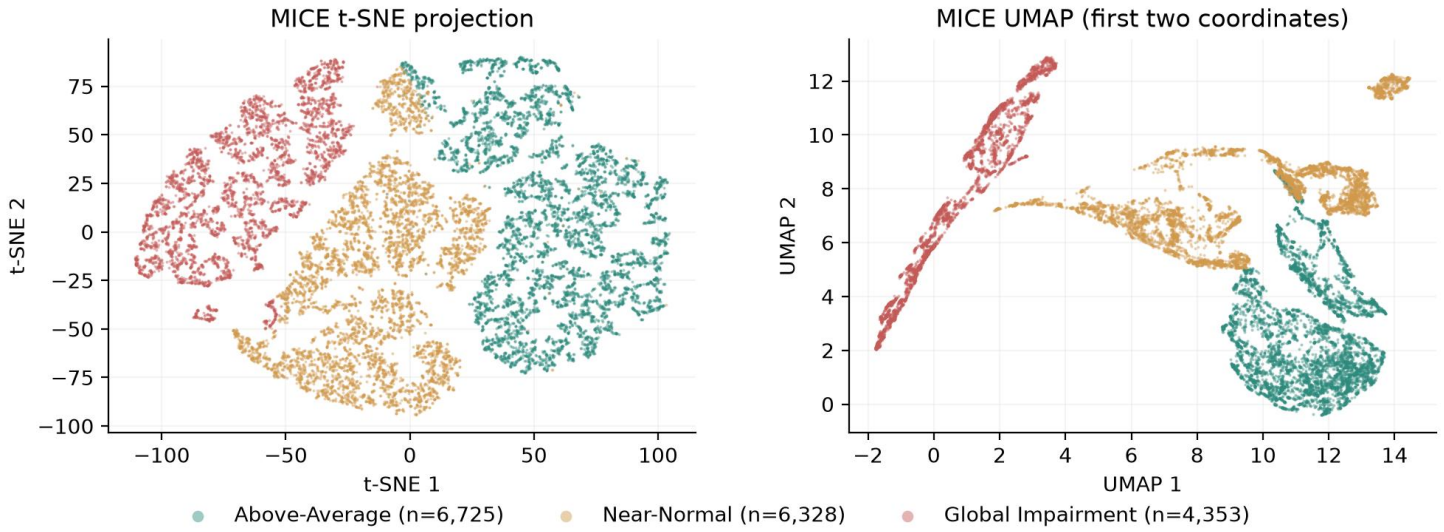
**Donor / matrix / deep learning**

6. Predictive mean matching
7. SoftImpute
8. Non-negative matrix factorization
9. Denoising autoencoder
10. Variational autoencoder

*Agreement under a fixed rule tests imputation sensitivity. Independent refitting and missingness assumptions require separate checks.*

RESULT · H1

# Three cognitive severity tiers emerge




**Above-Average**

above cohort mean

**6,725**

**Near-Normal**

close to the average

**6,328**

**Global Impairment**

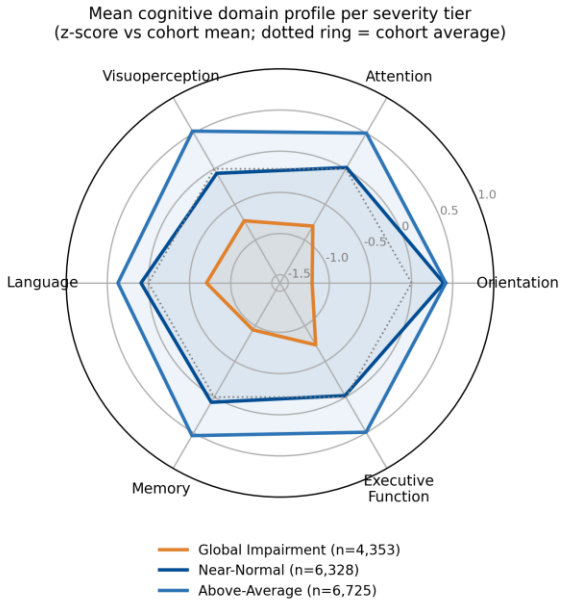
below cohort mean

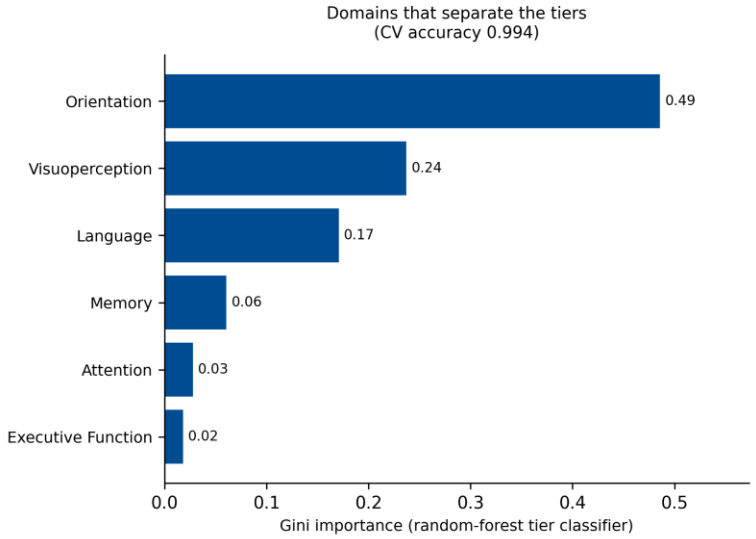
**4,353**

*All 10 methods meet the selected embedding thresholds. MICE silhouette ≈ 0.53 after reassignment; k = 3. Discrete phenotypes remain unconfirmed.*

INTERPRETATION · H1

# Continuum evidence: level, not shape




**PC1**

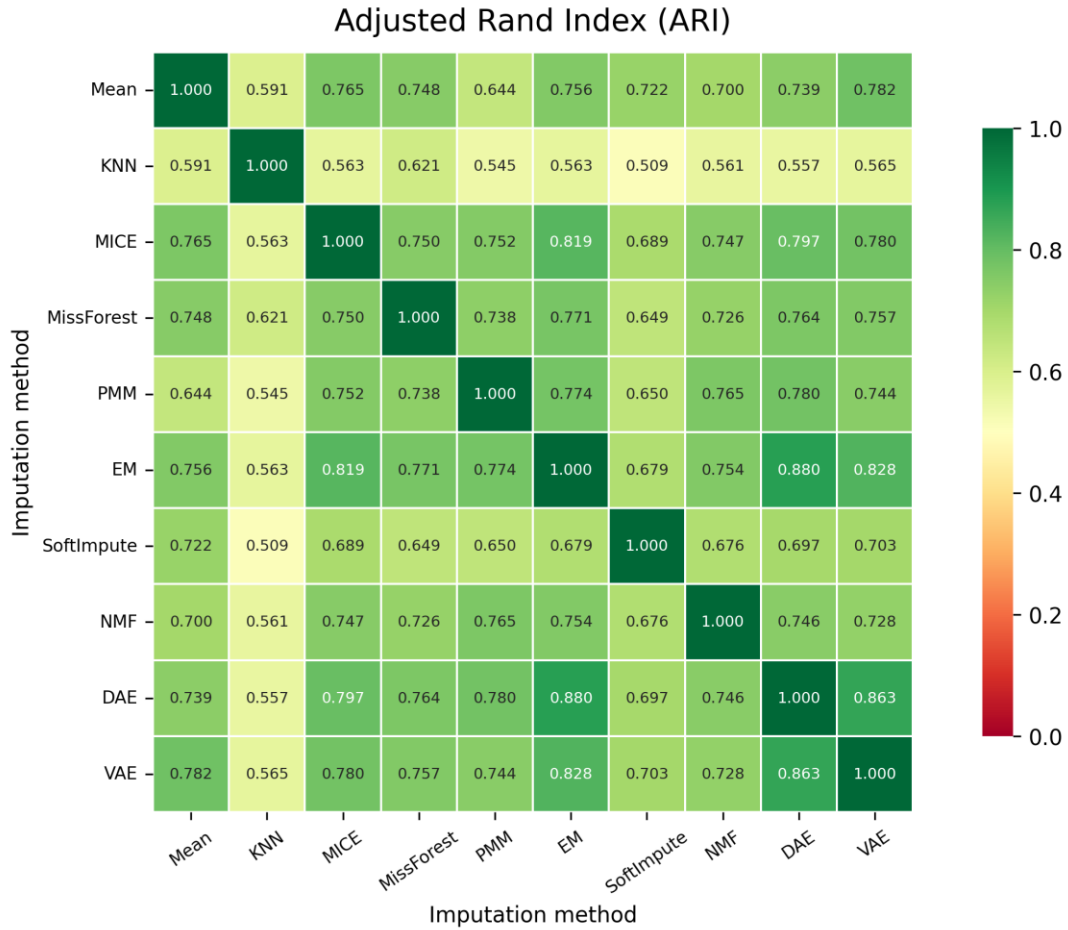
## ≈ 56%

of variance — one dominant axis

- Radar profiles are roughly concentric.
- All domains move in the same direction.
- All domains load positively on PC1.
- Signal = global severity, not dissociation.

RESULT · H2

# Robust across imputation methods




*Pairwise ARI across all ten imputation methods*

**0.71**

mean ARI · assessment-bootstrap 95% interval 0.702–0.717

**44 / 45**

method pairs survive Holm correction

**> 86%**

assessments with at least 80% assignment agreement

*Scope: point assignments are robust under a fixed scaler, embedding and partition — not full invariance of independently re-fit pipelines.*

RESULT · H2

# Complete-case analysis changes the answer

| | MICE reference | Complete-case |
|---|---|---|
| Assessments retained | 100% | 53.1% |
| Tiers recovered | 3 | 2 |
| Silhouette | ≈ 0.53 | ≈ 0.26 |
| ARI vs. reference (shared rows) | — | ≈ 0.63 |

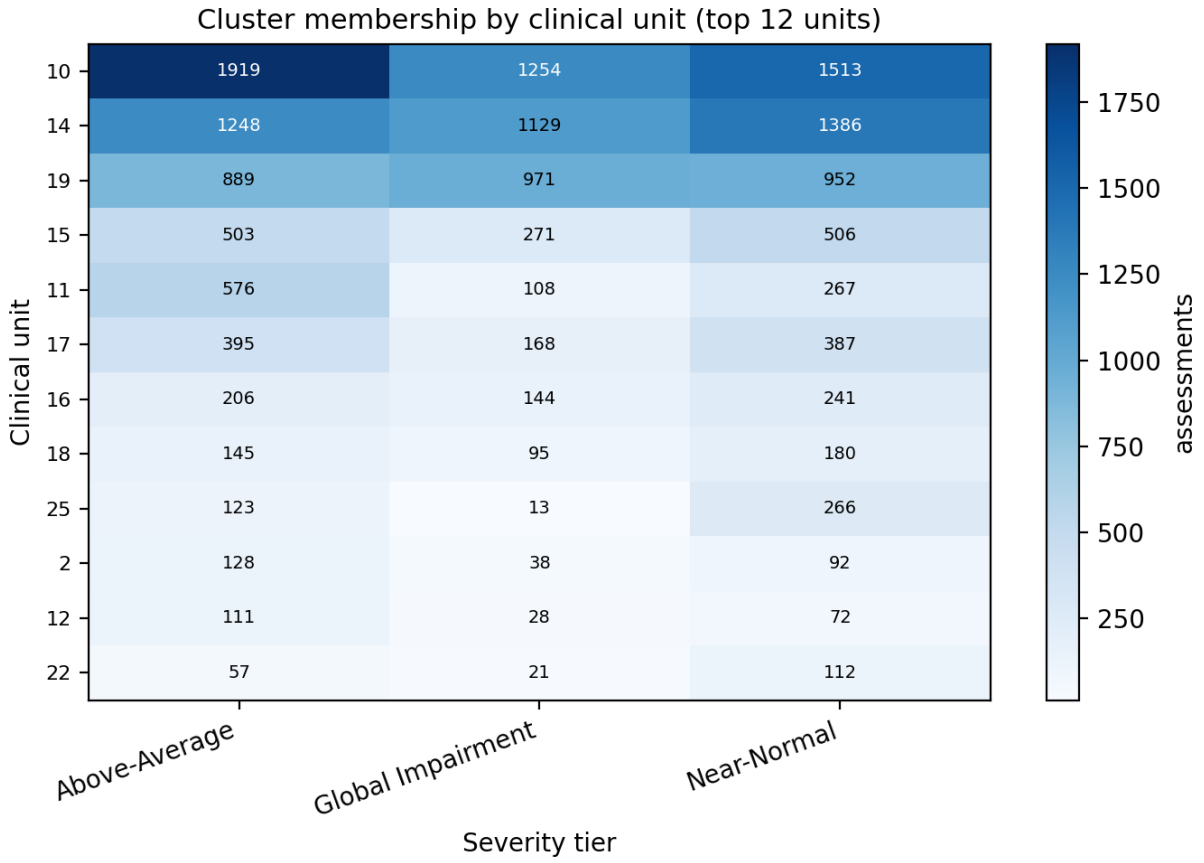
**Complete-case selection changes**

**the cohort and partition.**

The supplementary baseline used raw-test aggregation. These statistics do not validate the standardized patient-level pipeline.

RESULT · H3

# Tiers track clinical unit assignment




$\chi^2 = 911$ $p < 10^{-156}$

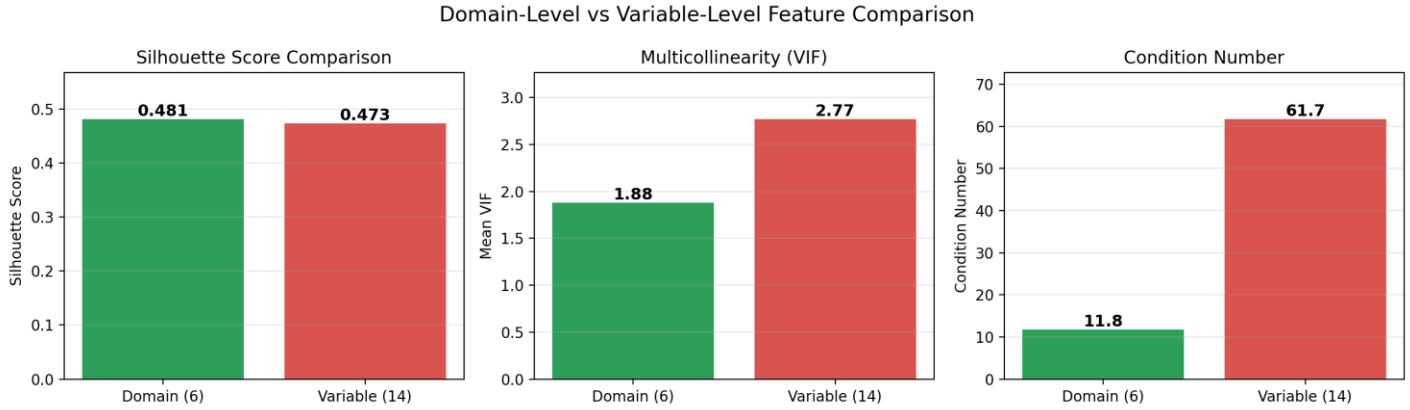
**0.16**

Cramér's V · criterion > 0.10

- Small association with clinical-unit assignment.
- Patient-grouped log-loss gain: 0.0013; interval includes zero.
- Added predictive value beyond diagnosis is unconfirmed.

RESULT · H4

# Domain aggregation improves conditioning




### Silhouette

~~0.473~~ → **0.481**

variables → domains

### Mean VIF

~~2.77~~ → **1.88**

variables → domains

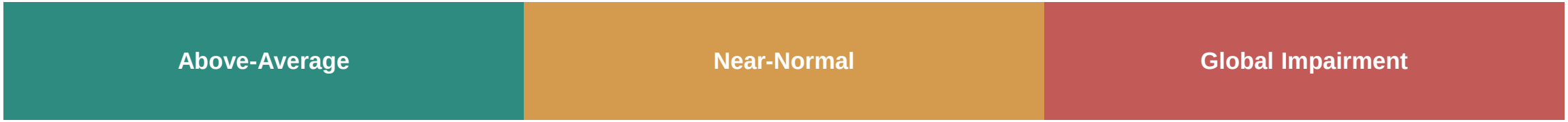
### Condition number

~~61.7~~ → **11.8**

variables → domains

WHAT IT MEANS

# A continuum, not a catalogue




- Subtypes would imply qualitatively different cognitive profiles.
- A gradient implies differences lie mostly along one severity axis.
- The tiers are practical bands extracted from that continuum.
- Domain-specific dissociations may exist — but they don't dominate here.
- Poor scaling, redundant or non-cognitive features, or complete-case selection can fake subtype structure.

IMPLICATIONS

# Clinical and analytic utility

## Clinical users

- Descriptive cognitive-score bands.
- Agreement scores express model consensus.
- Clinical utility still requires independent validation.

## Analytic users

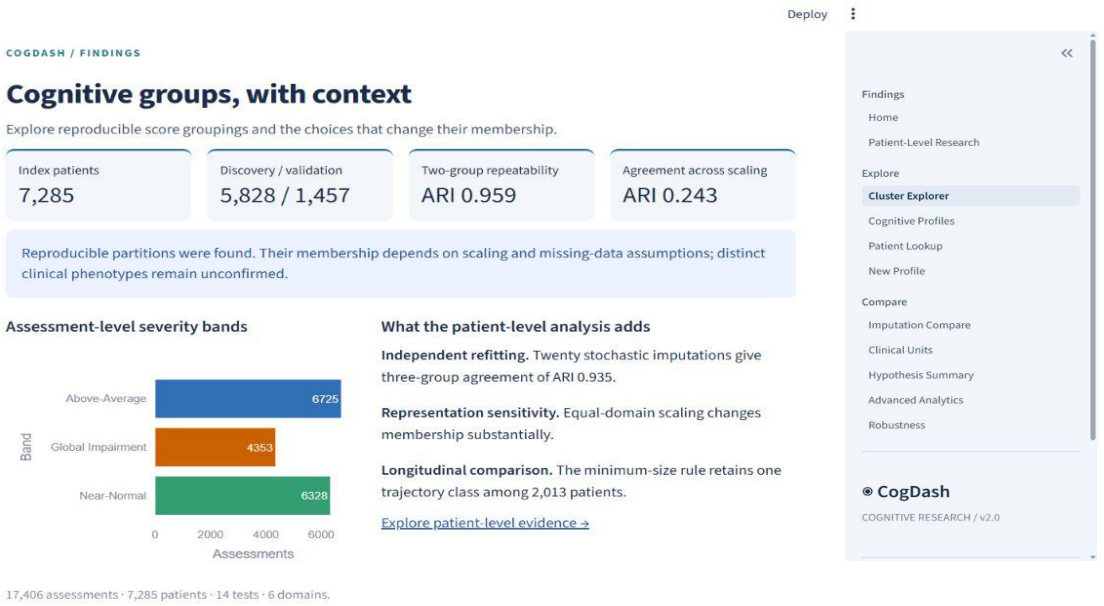
- Reproducible missingness-aware pipeline.
- Explicit, testable imputation decisions.
- Adaptable for linking cognition to outcomes.

*Long-term contribution: make missing-data decisions explicit, then test their influence — rather than hiding them.*

TRANSLATION

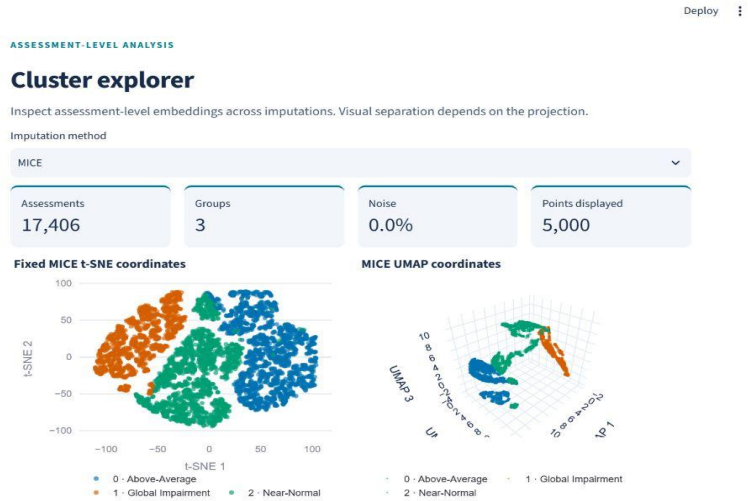
# CogDash — exploring the research findings










- Assessment-level and patient-level findings in eleven pages.
- Saved-model profile comparison reports distances, not clinical probabilities.
- Research use; diagnostic and treatment utility remain unconfirmed.

CAVEATS

# Limitations

### Single site

One rehabilitation centre — external replication is required.

### Longitudinal selection

One trajectory class meets the 5% size rule; nonlinear change and informative dropout remain unresolved.

### Missing-data assumptions

MAR is a working assumption; MNAR cannot be excluded.

### No functional outcomes

Return-to-work, length-of-stay and independence were unavailable.

### Embedding-based validation

UMAP-space silhouette may overstate separation.

### Pipeline sensitivity

Scaling changes patient-level membership substantially (ARI 0.243).

IN SUMMARY

# Three conclusions

| | | |
|---|---|---|
| 1 | **Empirical** | Repeatable score groups exist under specified choices. Distinct clinical phenotypes remain unconfirmed. |
| 2 | **Methodological** | Scaling, imputation and test availability materially affect group membership. |
| 3 | **Practical** | Domain aggregation improves conditioning. Clinical utility requires independent outcomes and external replication. |

THANK YOU

# Questions & discussion

*Reproducible cognitive score groupings, with sensitivity to scaling and missingness. Clinical phenotypes remain unconfirmed.*

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APPENDIX

# Backup slides for examiner questions

PATIENT-LEVEL EXTENSION

# Independent refitting and sensitivity

1. 5,828 discovery patients · 1,457 validation patients
2. 20 independent imputations: ARI 0.959 (2 groups), 0.935 (3 groups)
3. Robust versus equal-domain scaling: ARI 0.243
4. Student t mixtures select 8 components at the search boundary
5. 2,013 longitudinal patients: one class meets the 5% size rule

# Imputation hyperparameters

| Method | Hyperparameters |
|---|---|
| **Mean** | Single-pass arithmetic mean per variable. |
| **EM** | Multivariate normal model; convergence 1e−6; max 500 iterations. |
| **MICE** | Linear regression per variable; 10 iterations; single imputed dataset for clustering. |
| **KNN** | k = 5; inverse-distance weighting; Euclidean distance. |
| **MissForest** | 100 trees per forest; up to 10 iterations; variables imputed by increasing missingness. |
| **PMM** | 5 donors; 10 iterations; implemented within the MICE framework. |
| **SoftImpute** | Regularisation chosen by CV over 20 log-spaced values (1e−3 to 1e2); convergence 1e−5; max 200 iterations. |
| **NMF** | Rank r = 5 selected by grid search over {2, 3, 5, 7, 10}; multiplicative updates; max 500 iterations. |
| **DAE** | 15-128-64-32 autoencoder; ReLU + batch norm; dropout 0.5/0.3; 20% pattern-aware masking; MSE on observed entries; Adam (lr 1e−3); early stopping; 5 runs averaged. |
| **VAE** | Same layer dimensions as DAE; latent dim 32; loss = MSE + β-weighted KL (cyclical annealing, βmax 0.1); 20 latent samples averaged. |

APPENDIX · B

# H2 fixed-reference design

- Fit the reference scaler on MICE-imputed domain scores.
- Fit one UMAP embedding from the MICE reference.
- Fit a three-group k-means assignment rule.
- Apply the frozen transformations to other imputations.
- Compare assignments using pairwise ARI and NMI.

**Tests**

Robustness of point assignments to imputed values.

**Does not test**

Full invariance of independently discovered structures.

# HDBSCAN logic

- Estimate local density via core distance.
- Construct mutual-reachability distances.
- Build a minimum spanning tree.
- Extract a hierarchy of density-connected components.
- Select persistent clusters by stability.
- Label points outside persistent regions as noise.

*Unlike DBSCAN, HDBSCAN does not reduce to a single fixed-ε DBSCAN solution — it integrates across density levels.*

APPENDIX · D

# Silhouette caveats

- UMAP emphasizes neighbourhood structure.
- Embedding silhouette can overstate measured-domain separation.
- Hyperparameter and seed selection make the chosen score optimistic.
- Historical assessment subsampling gives mean ARI 0.273.

**Interpret silhouette as an internal pipeline-quality metric — not proof of natural categories.**

APPENDIX · E

# Missing-not-at-random sensitivity

- MAR is a working assumption.
- Observed data alone cannot rule out MNAR mechanisms.
- Agreement across imputations does not establish MNAR robustness.
- Clinically justified delta-shift sensitivity remains future work.

# Clinical interpretation boundary

### Research summaries provide

- cohort-relative score bands;
- assignment agreement across methods;
- exploratory profile comparison;
- candidates for outcome validation.

### The severity tier does NOT

- diagnose a patient;
- prescribe treatment;
- determine clinical placement;
- replace professional judgement.

# Future work

- External replication across rehabilitation centres.
- Nonlinear trajectories and informative follow-up or dropout.
- Validation against independent functional outcomes.
- Broader incomplete-data models and MNAR sensitivity.
- Prospective evaluation of CogDash before clinical use.
- Continuous scores versus added value from group membership.

# How to read the metrics

## Clustering quality & robustness

### Silhouette

Cohesion vs. separation — how well a point fits its own tier vs. the nearest other.

**−1 to 1 · criterion ≥ 0.40 · selected embedding ≈ 0.53**

### Adjusted Rand Index (ARI)

Agreement between two partitions, corrected for chance.

**0 = chance, 1 = identical · median criterion > 0.50**

### Normalised Mutual Information

Shared information between two partitions, scaled 0–1.

**0 = independent, 1 = identical · corroborates ARI**

### Consensus agreement

For each assessment, the fraction of methods assigning its modal label.

**86.2% of assessments have agreement ≥ 80%**

## Association & conditioning

### $\chi^2$ statistic

Evidence that tier and clinical unit are associated — significance, not effect size; grows with N.

**$\chi^2 = 911$ · $p < 10^{-156}$**

### Cramér's V

Effect size of that association, scaled 0–1.

**Interpret with table dimensions · observed 0.162**

### VIF (variance inflation factor)

Per-feature redundancy from collinearity.

**1 = none; > 5–10 concerning · 2.77 → 1.88**

### Condition number

Global conditioning of the feature set (λmax / λmin of the correlation matrix).

**Smaller is better conditioned · 61.7 → 11.8**